Detailed Calculations: Conceptual Greywater Bioreactor

Calculations for Previous Hybrid Algal Bioreactor System Design

IMPORTANT NOTE: These calculations pertain specifically to the previous conceptual design of the Hybrid Algal Bioreactor System, which featured separate Flat Panel Photobioreactors and a distinct Airlift Column.

For the refined design, new calculations for bioreactor volume distribution, component dimensions, and pump sizing are required, as the physical configuration and integration are fundamentally different. These calculations are provided here for context and to show the evolution of the design process.

(For 1 Household / 4 People in Low-Income Indian Context)

Phase 1: Bioreactor Volume Calculations

1. Calculate Total Bioreactor Volume (V) based on Hydraulic Retention Time (HRT): This total volume ensures the greywater stays in the bioreactor for the desired treatment period (HRT) for the specified daily flow.

Formula: VTotal =Daily Greywater Flow (Q)×HRT

Calculation (using Q = 144 L/day and HRT = 6.5 days):

VTotal =144 L/day×6.5 days VTotal =936 Liters

2. Distribute Total Volume between Flat Panel Photobioreactors (PBRs) and Airlift Column:

Assumption: Flat panels contribute approximately 90% of the total working volume, and the Airlift Column contributes the remaining 10%.

Calculations:

Target Volume for Flat Panels: 936 L×0.90=842.4 Liters

Target Volume for Airlift Column: 936 L×0.10=93.6 Liters

3. Calculate Dimensions and Number of Flat Panel Photobioreactors (PBRs): * Assumed Dimensions for a Single Flat Panel:

Width (W): 0.15 m (15 cm) - *Reasoning: Narrow width for optimal light penetration.*

Height (H): 2.5 m - *Reasoning: Balances vertical space utilization with structural stability and access.*

Length (L): 1.2 m - *Reasoning: Creates a manageable module size for manufacturing, transport, and installation.*

Calculate Volume of a Single Flat Panel:

Formula: Volume (m3) = W (m) × H (m) × L (m)

Calculation: 0.15 m×2.5 m×1.2 m=0.450 m3=450 Liters

Calculate Total Number of Flat Panels Required:

Formula: Number of Panels=Volume of a Single PBR Panel Total Target PBR Volume

Calculation: 450 L/panel842.4 L =1.872 panels≈2 panels

Initial Footprint Reflection for PBRs:

Decision: Only 2 Flat Panels would be required for a single household. This is a highly practical and compact number, aligning with the project's goal of a 3-4 panel system (allowing for flexibility or a slightly larger design if needed).

4. Calculate Dimensions for the Airlift Column(s):

Assumed Number of Airlift Columns: 1 central airlift column.

Assumed Height (H) for the single airlift column: 2.0 m - *Reasoning: Provides sufficient height for efficient mixing and gas exchange, while remaining practical for household installation.*

Calculate Required Diameter (D) for the single Airlift Column:

Formula: D=2×π×HVolume (m3)

Calculation (using Target Volume = 93.6 L = 0.0936 m^3): D=2×3.14159×2.0 m0.0936 m3   D=2×6.283180.0936   D=2×0.014896 D≈2×0.122 m=0.244 meters or 24.4 cm

Dimensions: The single airlift column would have a diameter of 0.244 meters (24.4 cm) and a height of 2.0 meters.

5. Consolidate & Verify Total System Volume:

Total Calculated System Volume: Sum of Flat Panel Volume (842.4 L) + Sum of Airlift Column Volume (93.6 L) = 936 Liters

Verification: This total calculated volume of 936 Liters precisely matches the initial target total bioreactor volume (from HRT calculation). This confirms the validity of these conceptual dimensions for a single household system.

Phase 2: Nutrient Removal & Biomass Production Calculations

6. Calculate Daily Mass of Phosphorus (P) to be Removed: This is the total amount of phosphorus that needs to be assimilated by the algae daily.

Formula: P to remove=Pin×Q×Target Removal Efficiency

Calculation: P to remove=0.330 mg/L×144 L/day×0.95 P to remove=45.144 mg P/day

7. Calculate Daily Mass of Nitrogen (N) to be Removed: This is the total amount of nitrogen that needs to be assimilated by the algae daily.

Formula: N to remove=N-in×Q×Target Removal Efficiency

Calculation: N to remove=18.98 mg/L×144 L/day×0.95 N to remove=2594.16 mg N/day

8. Calculate Daily Dry Biomass Required based on Phosphorus Removal (Biomass-P): This calculates how much dry algae biomass needs to be produced to remove the target amount of phosphorus.

Formula: Biomass-P=P Content in Dry Algae Biomass P to remove

Calculation: Biomass-P (mg)=0.0145.144 mg P/day Biomass-P (mg)=4514.4 mg dry algae/day *Convert to kg:* Biomass-P (kg)=1,000,000 mg/kg4514.4 mg =0.00451 kg dry algae/day

9. Calculate Daily Dry Biomass Required Based on Nitrogen (N) Removal (Biomass-N): This calculates how much dry algae biomass needs to be produced to remove the target amount of nitrogen.

Formula: Biomass-N=N Content in Dry Algae Biomass N to remove

Calculation: Biomass-N (mg)=0.082594.16 mg N/day Biomass-N (mg)=32427 mg dry algae/day *Convert to kg:* Biomass-N (kg)=1,000,000 mg/kg32427 mg =0.03243 kg dry algae/day

10. Determine Final Expected Daily Dry Chlorella vulgaris Biomass Production: The system must produce enough biomass to remove *both* nutrients. Therefore, we take the larger of the two biomass requirements.

Comparison: 0.00451 kg/day (from P removal) vs. 0.03243 kg/day (from N removal)

Decision: The nitrogen removal dictates the biomass production.

Final Daily Biomass Production: 0.03243 kg dry algae/day

Phase 3: CO_2 Capture Calculation

11. Calculate Mass of Carbon Contained in Algae Biomass: This determines how much carbon is locked away in the produced algae biomass.

Formula: Mass of Carbon=Final Daily Biomass Production×Algae Carbon Content

Calculation: Mass of Carbon=0.03243 kg dry algae/day×0.50 Mass of Carbon=0.016215 kg C/day

12. Calculate Mass of CO_2 Captured: This converts the mass of carbon in the algae to the equivalent mass of CO_2 captured from the atmosphere (using molecular weights: C=12, O=16, so CO_2=44).

Formula: Mass of CO2 captured=Mass of Carbon×Molecular Weight of CMolecular Weight of CO2   Calculation: Mass of CO2 captured=0.016215 kg C/day×1244 Mass of CO2 captured=0.059455 kg CO2 /day

Summary of Final Calculated Outputs:

- Final Target Bioreactor Volume: 936 Liters
- Number of Flat Panel Photobioreactors: 2 panels
- Airlift Column Dimensions: Diameter 0.244 m (24.4 cm), Height 2.0 m
- Final Daily Dry *Chlorella vulgaris* Biomass Production: 0.03243 kg/day
- Daily CO_2 Capture Potential: 0.059455 kg CO_2/day